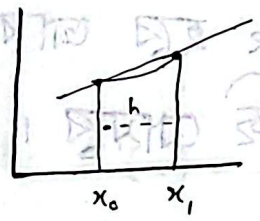
⊛ L10 : Numerical Differentiation

[Ch-3.1]

Actual differentiation: $\frac{d}{dx}(x^2-5x) = 2x-5$ ← computer এভাবে differentiate করতে পারে না

* $x_1 = x_0 + h$



Formula

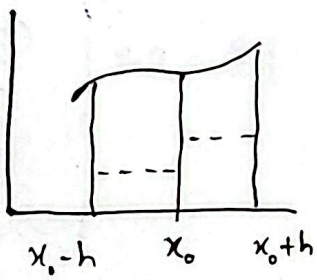
Forward Differentiation,

$$f'(x) = \frac{f(x_0+h)-f(x_0)}{h} + \left|\frac{f''(\varepsilon)(h)}{2}\right|$$

Numerical / Forward differentiation

Truncation error of Numeric Differentiation / Forward differentiation

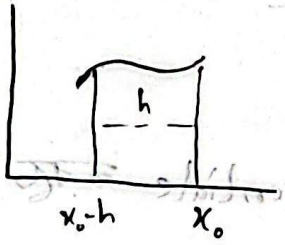
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Central Differentiation,

$$f(x) = \frac{f(x_0+h)-f(x_0-h)}{2h} + \left|\frac{f'''(\varepsilon_.)h^2}{3!}\right|$$

Backward differentiation,



$$f'(x) = \frac{f(x_0) - f(x_0-h)}{h} + \left| \frac{f''(\varepsilon)\, h}{2} \right|$$

(h: step size)

* h যত ছোট হবে তত বেশি accurate হবে আর actual differentiation এর কাছাকাছি আসবে।

$f(x) = 2x^2$. $f'(2) = ?$

(a) Find actual differentiation
(b) If $h = 0.1$. Find forward differentiation.
(c) Truncation error from actual error.

Ans:

(a) Actual differentiation •,

$f'(x) = 4x$

$f'(2) = 8$

(b) $f'(x) = \frac{f(x_0+h) - f(x_0)}{h}$

$= \frac{2(x_0+h)^2 - 2x_0^2}{h}$

$f'(2) = \frac{2(2+0.1) - 2 \cdot 2^2}{0.1}$

$= 8.2$

(c) $|8.2 - 8| = 0.2$

$f(x) = e^x - 2x$; $h = 0.1$

$f'(2) = ?$ Use central differentiation

Ans:

Central differentiation, $f'(x_0) = \dfrac{f(x_0+h) - f(x_0-h)}{2h}$

$$= \frac{e^{x_0+h} - 2(x_0+h) - e^{x_0-h} + 2(x_0-h)}{2h}$$

$$= \frac{e^{2+0.1} - 2(2+0.1) - e^{2-0.1} + 2(2-0.1)}{2 \times 0.1}$$

$$= 5.401$$

Actual differentiation, $f(x) = e^x - 2x$

$f'(x) = e^x - 2$

$f'(2) = e^2 - 2$

$= 5.389$

x	f(x)
4	16
4.1	18
4.2	20
4.3	21
4.4	22

(0.1 between 4 and 4.1; 0.1 between 4.1 and 4.2; 0.1 between 4.2 and 4.3; 0.1 between 4.3 and 4.4)

Backward $\therefore f'(4.2)$

Ans:

$$f'(x) = \frac{f(x) - f(x-h)}{h} = \frac{f(4.2) - f(4.2-0.1)}{0.1}$$

$$= \frac{20 - 18}{0.1}$$

$n = 2$, $f(x) = \cos(x)$, $\varepsilon \in [1, 1.5]$

$h = 0.1$

Find Truncation error / Upper bound error for Forward differentiation

Ans:

$$\left| \frac{f''(\varepsilon)(-h)}{2} \right| = \left| -\cos(\varepsilon) \right|_{\substack{\text{Max} \\ \varepsilon \in [1,1.5]}} \left| \frac{-h}{2} \right|$$

$f(x) = \cos x$

$f'(x) = -\sin x$

$f''(x) = -\cos x$

$$= \left| -\cos(\varepsilon) \right|_{\substack{\text{Max} \\ \varepsilon \in [1,1.5]}} \left| \frac{-h}{2} \right|$$

$\cos(1) = 0.54$ ← Max

$\cos(1.2) = 0.36$

$\cos(1.5) = 0.07$

$$= 0.54 \times 0.05$$

$$= 0.027$$

$f(x) = xe^{-3x} + x^2$

Calculate the upper bound truncation error at $x_0 = 2$ using $h = 0.1$ for central differentiation.

Ans:

$$\text{Truncation error} = \left| \frac{f'''(\varepsilon)\, h^2}{3!} \right|$$

Now,

$$f(x) = xe^{-3x} + x^2$$

$$f'(x) = e^{-3x} - 3xe^{-3x} + 2x$$

$$= (1-3x)e^{-3x} + 2x$$

$$f''(x) = (9x-6)e^{-3x} + 2$$

$$f'''(x) = \frac{d}{dx}(9xe^{-3x}) - \frac{d}{dx}(6e^{-3x})$$

$$= 9e^{-3x} + 9x \cdot (-3)e^{-3x} - 18e^{-3x}$$

$$= 9e^{-3x} - 27xe^{-3x} - 18e^{-3x}$$

$$= 27(1-x)e^{-3x}$$

Central ε এর value নেওয়া হয়

~~Forward: $\varepsilon \in [x_0, x_0+h]$~~
~~Backward: $\varepsilon \in [x_0-h, x_0]$~~

$\varepsilon \in [x_0-h, x_0+h]$

$\Rightarrow \varepsilon \in [1.9, 2.1]$

Forward: $\varepsilon \in [x_0, x_0+h]$
Backward: $\varepsilon \in [x_0-h, x_0]$
Central: $\varepsilon \in [x_0-h, x_0+h]$

Now,

$$\text{Truncation error} = \left| 27(1-\varepsilon)e^{-3(\varepsilon)} \right| \left| \frac{0.1^2}{3!} \right|$$

$$= 27 \times |(1-\varepsilon)| \times \left| e^{-3(\varepsilon)} \right| \times \frac{1}{600}$$

$$= 27 \times 3.34 \times 10^{-3} \times 1.1 \times \frac{1}{600}$$

$$= 1.35 \times 10^{-4}$$

Rough:

If $\varepsilon = 1.9$: • $(1-1.9) = |-0.9|$
• $e^{-3(1.9)} = 3.34 \times 10^{-3}$ [Max]

$\varepsilon = 2$: • $(1-2) = |-1|$
• $e^{-3(2)} = 2.48 \times 10^{-3}$

$\varepsilon = 2.1$: • $(1-2.1) = |-1.1|$ [Max]
• $e^{-3(2.1)} = 1.83 \times 10^{-3}$

$\varepsilon = \frac{1.9+2.1}{2}$